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CLOPE IN R

TRANSACTION
CLUSTERING FOR
MESSY,
IRREGULAR DATA



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The Problem

Most clustering algorithms struggle with:

- messy categorical data
- irregular structure
- unknown number of clusters
- scaling beyond ~10k rows

K-Means falls over here.

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The Alternative

CLOPE

A fast, scalable clustering algorithm
for **transaction-style data**.

No distance matrix.

No need to pre-specify cluster
count.

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Where I first used it

~10 years ago: SAP work-order text.
No R implementation existed.
Switching tools broke the workflow.

So I built it.

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Where I use it now

Clustering file headers across large collections of CSVs:

- assay files
- collar files
- survey files
- geometallurgy testwork

CLOPE reveals structure instantly.

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Why this matters

CLOPE is part of my data profiling workflows:

- data quality scoring (DQS)
- data discovery
- dataset classification
- automated ingestion

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What you can use it for

- geology summary logs
- mixed assay + geology + geotech datasets
- geometallurgy response data
- operational logs
- any dataset you can tokenise

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The R implementation

- **First attempt:** loop-based, slow, literal translation of the paper.
- **Idiomatic R:** refined, vectorised R, faster.

Now included in mining-ds-toolkit.

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mining-ds-toolkit

Folder: library/clope

Source: clope.R

